

EXPLORATORY DATA ANALYSIS ON LAPTOP DATA

This document outlines the exploratory data analysis conducted on a laptop dataset to understand key features, relationships, and trends related to laptop pricing and specifications.

1. Data Preparation:

- Added a unique id column as the primary key.
- Extracted first rows, last rows, and random samples for an overview.

2. Univariate Statistics:

- **Numerical Columns:**
 - Calculated basic statistics for numerical columns, including count, min, max, average, and standard deviation.
 - Computed quartiles (Q1, Median, Q3) for the Price column.
 - Checked for missing values and confirmed no null values in Price.
 - Identified outliers in Price using the IQR method.
 - Created a histogram to group `Price` into buckets and calculated frequency buckets.
- **Categorical Columns:**
 - Counted occurrences and calculated percentages for Company, TypeName, Touchscreen, MemoryType, and gpu_brand.
 - Identified the mode for PrimaryStorage and examined SecondaryStorage.

3. Bivariate Analysis:

- **Numerical - Numerical:**
 - Explored relationships like Price vs Weight, Price vs Ram, Price vs Inches, Price vs cpu_speed and Price vs PPI (using ResolutionWidth/Height) and calculated average Price for each.
- **Categorical - Categorical:**
 - Analysed combinations such as Company vs gpu_brand, Company vs TypeName, Company vs MemoryType, Company vs cpu_brand, cpu_brand vs gpu_brand by creating contingency tables.
- **Numerical - Categorical:**
 - Compared statistical measures (min, max, avg Price) across categories like Company, gpu_brand, TypeName, cpu_brand and OpSys.

4. Insights & Visualizations:

- Highlighted trends using charts such as stacked bar charts, 100% stacked column charts, scatter plots, clustered bar charts and etc.